Explanation of numbering system: The first one or two digits before the period refer to the textbook chapter to which the question pertains. The digits after the period refer to the number of the Test Bank question pertaining to the designated chapter. Thus, "3.1" refers to the first question pertaining to Chapter 3.

The quiz and final exam questions will be similar are style the questions found in this Test Bank.

Note that the default assumption in this course is that interest rates and dividend yields are assumed to be quoted on a per annum and continuously compounded basis.

Chapter 1: Introduction

1.1. A trader enters into a one-year short forward contract to sell an asset for $60 when the spot price is $58. The spot price in one year proves to be $63. What is the trader's profit?

Loss of $3

1.2. A trader buys 100 European call options with a strike price of $20 and a time to maturity of one year. Each option involves one unit of the underlying asset. The cost of each option or option premium is $2. The price of the underlying asset proves to be $25 in one year. What is the trader's profit?

Profit of $300

1.3. A trader sells 100 European put options with a strike price of $50 and a time to maturity of six months. Each option involves one unit of the underlying asset. The price received for each option is $4. The price of the underlying asset is $41 in six months. What is the trader's profit?

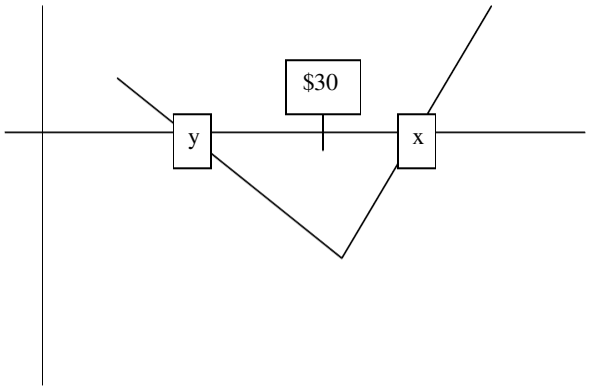
Loss of $500

1.4. The price of a stock is $36 and the price of a 3-month call option on the stock with a strike price of $36 is $3.60. Suppose a trader has $3,600 to invest and is trying to choose between buying 1,000 options and 100 shares of stock. How high does the stock price have to rise for an investment in options to be as profitable as an investment in the stock?

$40 Note that we are trying to solve the following equation for P, the stock price:
(P-36)100 = (P-36)1000 -3,600

1.5. A one year call option on a stock with a strike price of $30 costs $3. A one year put option on the stock with a strike price of $30 costs $4. A trader buys two call options and one put option.
A.) What is the breakeven stock price, above which the trader makes a profit?
B.) What is the breakeven stock price below which the trader makes a profit?

A.) $35 since 2=10/x' where x' is the amount by which the breakeven price exceeds $30, the strike price. Note that x = 30 + x'.
B.) $20 since 1=10/y' where y' is the amount by which the breakeven price falls short of $30, the strike price. Note that y = 30 – y'.



Chapter 2: Mechanics of Futures Markets

2.1. A company enters into a short futures contract that involves 50,000 pounds of cotton for 70 cents per pound. The initial margin is $4,000 and the maintenance margin is $3,000. What is the futures price above which there will be a margin call?

$0.72 since we are trying to solve the equation: ($.70-P) 50,000 = - $(4,000-3,000)

2.2. A company enters into a long futures contract involving 1,000 barrels of oil for $20 per barrel. The initial margin is $6,000 and the maintenance margin is $4,000. What oil futures price will allow $2,000 to be withdrawn from the margin account?

$22 since we are trying to solve the equation: 1000(P-20) = $2,000

Note that an amount can be withdrawn from the margin account when P, the settlement price on the day of the transaction of the oil futures contract, exceeds $20.

2.3. On the floor of a futures exchange one futures contract is traded where both the long and short parties are closing out existing positions. What is the resultant change in the open interest?

Open interest drops by one.

2.4. You sell 3 December gold futures when the futures price is $410 per ounce. Each contract is on 100 ounces of gold and the initial margin per contract is $2,000. The maintenance margin per contract is $1,500. During the next 7 days the futures price rises steadily to $412 per ounce. What is the balance of your margin account at the end of the 7 days?

$5,400 since the total initial margin of 3X$2,000 is reduced by 3X$(412-410)X100=$600

2.5. A hedger takes a long position in an oil futures contract on November 1, 2009 to hedge an exposure on March 1, 2010. Each contract is on 1,000 barrels of oil. The initial futures price is $20. On December 31, 2009 the futures price is $21 and on March 1, 2010 it is $24. The contract is closed out on March 1, 2010. What gain is recognized in the accounting year January 1 to December 31, 2010?

$4,000 = 1000 X $(24-20)

2.6. Answer 2.5 this time assuming that the trader in question is a speculator rather than a hedger.

$3,000 = 1000 X ($24-21)

2.7. A speculator enters into two short cotton futures contracts, when the futures price is $1.20 per pound. The contract entails the delivery of 50,000 pounds of cotton. The initial margin is $7,000 per contract and the maintenance margin is $5,250 per contract. The settlement price on the day of the transaction is $1.50 per pound. Assume that all days are trading days.

Notes:
1.) If there is a margin call on a certain day, the deadline for depositing the variation margin (which is the additional margin that should be deposited into the margin account due to a margin call) is the trading day after the day of the margin call. The assumption made in this course is that the variation margin is deposited at the deadline date, i.e. the trading day after the day of the margin call.
2.) Margin calls are established at the settlement price, i.e. margin calls are established at the end of the trading day.

A.) How much must the speculator deposit into his margin account on the day of the transaction?

Initial margin = 2 x $7,000 = $14,000

B.) What is the amount of the margin call, if any, that is declared on the day of the transaction?

Automatic credit to MAB (margin account balance) due to adverse move in the futures price, i.e., transaction price of 1.20 is less than the settlement price of 1.50, = 2 x 50,000 x (1.20 – 1.50) = -$30,000. A negative credit is a debit, i.e., the MAB is reduced by $30,000.

The initial margin that is deposited of $14,000 is reduced by $30,000, resulting in a MAB of -$16,000. As the latter is below the maintenance margin of $5,250 x 2 or $10,500, an additional deposit of $30,000 is required to bring the MAB back to the initial margin. The margin call thus equals $30,000.

C.) How much must the speculator deposit into his margin account, i.e. what is the variation margin, on the day after the transaction?

The margin call or variation margin of $30,000, calculated in B.), must be deposited. Note that margin calls or variation margins must be deposited on or before the trading day after the day of the margin call.

2.8. On a certain day a speculator enters into 10 long soybean futures contracts, when the futures price is $10.20 per bushel. The contract involves 5,000 bushels of soybean. The initial margin is $4,000 per contract and the maintenance margin is $3,000 per contract. The settlement price on that day is $10.05 per bushel. How much must the speculator deposit into his margin account on day 1?

Note: Quiz and exam questions will broach what transpires on only one trading day.

Initial margin = $4,000 x 10 = $40,000
Maintenance margin = $3,000 x 10 = $30,000
Automatic credit = 10 x 5,000 (10.05 – 10.20) = -7,500
Margin account balance = 40,000 – 7,500 = 32,500 which exceeds maintenance margin of 30,000. Thus, there is no variation margin required, i.e. there will be no margin call.
Deposit for day 1 = $40,000

2.9. List and explain briefly the possible effects of a single futures transaction on open interest.

Open interest rises by 1 if both long and short positions are opening transactions.

Open interest does not change if one of the long or short positions is an opening transactions whereas the other position is a closing transaction.

Open interest drops by 1 if both long and short positions are closing transactions.

Chapter 3: Hedging Strategies Using Futures

3.1. On March 1 the spot price of oil is $20 and the July futures price is $19. On June 1 the spot price of oil is $24 and the July futures price is $23.50. A company entered into a futures contract on March 1 to hedge the purchase of oil on June 1. It closed out the position on June 1. What is the effective price paid by the company for the oil?

$19.50 = $24 + $(19 - 23.50). Hedging involves adding a hedge $(19 – 23.50) to an initial exposure $24.

Alternatively, $19.50 = $19 + $(24 - 23.50). Hedging involves taking an initial futures position $19 and a basis $(24 – 23.50) that substitutes for the exposure.

3.2. On March 1 the spot price of gold is $300 and the December futures price is $315. On November 1 the spot price of gold is $280 and the December futures price is $281. A gold producer entered into a December futures contract on March 1 to hedge the sale of gold on November 1. It closed out its position on November 1. What is the effective price received by the producer for the gold?

$314 = $280 + $(315 - 281) or alternatively, $314 = $315 + $(280 – 281).

See 3.1 for the interpretations of these two equivalent calculations.

3.3. The standard deviation of monthly changes in the price of a commodity A is $2. The standard deviation of monthly changes in a futures price for a contract on commodity B, which is similar to commodity A, is $3. Note: This is an example of cross-hedging. The correlation between the futures price and the commodity price is 0.9.
A.) What hedge ratio should be used when hedging a one month exposure to the price of commodity A?

0.6 = .9 (2/3)

B.) What is the associated hedging effectiveness? Interpret what this means.

.81 = (.9)^2 The proportion of the variance of commodity A that can be eliminated by hedging with commodity B futures is 81%.

Note: A perfect hedge is one whose measure of hedging effectiveness is 100% or 1. This occurs when R^2 = 1. Alternatively, this occurs when the correlation between the changes in futures and spot prices equal 1.

3.4. A company has a $36 million portfolio with a beta of 1.2. The S&P 500 Index futures price currently equals 900. What trade in S&P Index Futures is necessary to achieve the following? Indicate the number of contracts that should be traded and whether the position is long or short.

A.) Eliminate all systematic risk in the portfolio.

Short 192 since N = (0-1.2) 36M/(900 x 250)= -192

Note: For S&P 500 Index futures contracts, F in the stock index formula equals 250xfutures price. For Mini S&P 500 Index futures contracts, F in the stock index formula equals 50xfutures price.

B.) Reduce the beta to 0.9.

Short 48 = (.9-1.2)36M/(900 x 250) = -48

C.) Increase beta to 1.8.

Long 96 = (1.8 – 1.2)36M/(900 x 250) = 96

3.5. The standard deviation of weekly changes in the spot price of pork bellies is 2.3 cents per pound. For pork belly futures that expire 6 weeks from now, the same standard deviation measures 3.9 cents per pound. The correlation between these two prices, i.e., spot and futures, is 0.65. Each pork belly futures contract entails the delivery of 20,000 pounds. A pork farmer is committed to delivering 100,000 pounds of pork bellies 4 weeks from now.
A.) What should the pork farmer do to hedge his exposure?

$N = \rho \frac{\sigma_S}{\sigma_F} \frac{Q_A}{Q_F} = (.65) \frac{2.3}{3.9} \frac{100{,}000}{20{,}000} = 1.9 \approx 2$ Applying the anticipatory hedging rule, to wit, do in the futures market now what you expect to do in the spot market in the future, the farmer should short 2 futures contracts.

Parenthetical Note: The standard deviation for a 4-week period equals $\sqrt{4}$ times the 1-week standard deviation. Observe that as the same constant term of $\sqrt{4}$ is present in both numerator and denominator of the ratio of standard deviations found in the formula, that constant term cancels out. Thus, the standard deviations employed in the formula may both be 1-week standard deviations rather than 4-week standard deviations.

B.) What percent of his exposure can the pork farmer eliminate by hedging?

$R^2 = \rho^2 = (.65)^2 = 42\%$ The farmer can eliminate 42% of his exposure by hedging, i.e., observing the advice offered in part A.).

3.6. An investment manager is in charge of a $55 million common stock portfolio whose beta equals 1.75. The S&P 500 Index futures price currently equals 1040.
A.) What should the manager do to hedge his portfolio using S&P 500 Index futures contracts?

$longN = \left(\beta^* - \beta\right)\frac{P}{F} = (0 - 1.75)\frac{55M}{.26M} = -370$ Thus, the manager must short 370 S&P 500 Index futures contracts. Note that F = 250 x 1040 = .26M, where M denotes a million.

B.)What should the manager do to increase the beta of his portfolio to a value of 2.2 using Mini S&P 500 Index futures contracts?

$longN = \left(\beta^* - \beta\right)\frac{P}{F} = (2.2 - 1.75)\frac{55M}{.052M} = 476$ Thus, the manager must take a long position in 476 Mini S&P 500 Index futures contracts. Note that F = 50 x 1040 = .052 M.

3.7. An agricultural cooperative would like to hedge the sale of one million bushels of grade 2 yellow corn that is scheduled to take place a month from now, employing CME corn futures contracts. The contract involves the delivery of 5,000 bushels of grade 1 yellow corn. The standard deviation of monthly changes in grade 2 yellow corn prices per bushel equals $2.30 while the standard deviation of monthly changes in grade 1 yellow corn futures prices per bushel equals $ 2.62. The correlation between these two prices equals 0.89. Presently, the price of grade 2 yellow corn per bushel equals $36.75 while the price of grade 1 yellow per bushel equals $38.95.
A.) (4%) What do you recommend that the agricultural cooperative do, ignoring the tailing the hedge adjustment?

$$h = \rho\frac{\sigma_S}{\sigma_F} = (.89)\frac{2.3}{2.62} = .7813$$
$$N = h\frac{Q_A}{Q_F} = (.7813)\frac{1M}{.005M} = 156$$

The cooperative should short 156 contracts.

Note that, in the absence of the phrase “ignoring the tailing the hedge adjustment,” you should take account of tailing the hedge. This is because you are hedging a future spot transaction with a futures contract.

Hedging a future transaction with a futures contract always requires that the hedge be tailed because of marking to market, i.e., the hedging activity generates immediate cash flows whereas the exposure pertains to a future event. Thus, tailing the hedge is a time value of money adjustment.

B.) (4%) What do you recommend that the agricultural cooperative do, taking account of the tailing the hedge adjustment?

$$N_{TH} = N\frac{S}{F} = (156)\frac{36.75}{38.95} = 147$$

Short 147 contracts.

Note that the textbook formula for Nth = h Va/Vf and the above formula are equivalent. This is because the quantity in the textbook formula numerator is Va = Qa x S and the quantity in the textbook formula denominator is Vf = Qf x F.

3.8. An investment manager, who is in charge of a $100 million stock portfolio with a beta of 1.5, projects that the stock market during the year that has just started will rise. He wishes to speculate on this belief. There are two actions he could take, namely, reduce the portfolio beta to 1 or raise it to 2. The S&P 500 Index futures price currently equals 1,200. What position should the investment manager take in Mini S&P 500 Index futures contracts?

$$LongN = \left(\beta^* - \beta\right)\frac{P}{F} = (2-1.5)\frac{100M}{.06M} = 833$$

$F = 50x1{,}200 = .06M$

Take long position in 833 contracts.

Chapter 4: Interest Rates

4.1. An interest rate is 15% per annum with annual compounding. What is the equivalent rate with continuous compounding?

13.98% since 1.15 = e^R implies R = 13.98%

4.2. An interest rate of 12% assumes quarterly compounding. What is the equivalent rate with semiannual compounding?

12.18% since (1 + 12%/4) = (1 + R/2)^2 implies R = 12.18%

4.3. A.) The 3-year zero rate is 7% and the 4-year zero rate is 7.5%, both continuously compounded. What is the forward rate for the fourth year?

9% =((7.5%)4 – (7%)3) / (4-3)

4.3 B.) For the situation depicted in part A.), what contractual interest rate would be appropriate for a one-year FRA that starts 3 years from now?

The continuously compounded forward rate of 9% must be restated as the equivalent forward rate with annual compounding, i.e. e^9% = (1+R). Thus, the contractual forward rate for the FRA is R = 9.42%.

Note that the quoted contractual forward rate of an FRA assumes a compounding period equal to the length of the FRA period. In this case, the FRA period is one year.

4.4. The 6-month zero rate is 8% with semiannual compounding. The price of a 1-year bond that provides a coupon of 6% per annum semiannually is 97. What is the one year zero rate continuously compounded?

9.02% since 3/1.04 + 103e^R = 97 implies R = 9.02%

The foregoing is a short problem on the bootstrapping procedure for generating the zero curve.

4.5. The zero curve is flat at 5.91% with continuous compounding. What is the value of an FRA to an FRA seller where the FRA interest rate is 8% per annum on a principal of $1,000 for a 6-month period that start 2 years from now?

Notes:
1.) FRA interest rates are quoted assuming a compounding period equal to the length of the FRA period. Thus, the 8% should be interpreted as semi-annually compounded.
2.) Since the zero curve is flat, all forward interest rates equals the constant value of the interest rate. Thus, the relevant forward rate is 5.91% continuously compounded or 6% with semi-annual compounding.
3.) This problem asks you to value the FRA post-inception. At inception, the value of an FRA equals 0.
4.) The seller of an FRA receives the contractual interest rate of the FRA. The seller is hedging a floating rate deposit.

$8.63 = 1000(.08-.06).5 x e^-5.91%(2.5)

or $8.63 = 1000(.08-.06).5 / (1.03)^5

4.6.A.) The 1-year spot (or zero) rate equals 5% and the 15-month spot rate equals 5.6%. What is the forward rate pertaining to the quarter that starts a year from now? All the interest rates cited here are expressed with continuous compounding.

$$R_F = \frac{5.6\%(1.25) - 5\%(1)}{(1.25 - 1)} = 8\%$$

4.6.B.) A firm, confronting the situation in part A.), wishes to purchase an FRA (Forward Rate Agreement) for the 1-quarter period that starts a year from now. What value of the contractual rate should the firm expect from a bank? By convention, the interest rates associated with an FRA assume a compounding period equal to the FRA's time period.

$$e^{8\%(.25)} = \left(1 + \frac{R_4}{4}\right) = 8.08\%$$

4.7. A 6-month T-bill is currently trading at $94. A 7% coupon rate 1-year maturity bond currently trades at $90. What are the 6-month and 1-year zero rates? All interest rates

cited here are continuously compounded. The bond is a traditional North American bond that pays coupons semi-annually.

$$94 = 100e^{-R_{0.5}(0.5)}$$
$$R_{0.5} = 12.375\%$$
$$90 = 3.5e^{-12.375\%(0.5)} + 103.5e^{R_1(1)}$$
$$R_1 = 17.7\%$$

4.8. A company has entered into an FRA (Forward Rate Agreement), which specifies that the company will receive 7%, quoted with semi-annual compounding, on a principal of $100 million for the 6-month period starting a year from now. The 1-year spot rate and the 18-month spot rate are 7% and 7.5%, respectively, both rates expressed as continuously compounded rates. What is the value of the company's FRA?

$$R_F = \frac{7.5\%(1.5) - 7\%(1)}{.5} = 8.5\%$$
$$e^{8.5\%(.5)} = (1 + \frac{R_2}{2})$$
$$R_2 = 8.68\%$$
$$V_{FRA} = \left[100Mx(7\% - 8.68\%)x.5\right]e^{-7.5\%(1.5)} = -\$750{,}622$$

4.9 A.) A 1-year maturity T-bill is trading at $94. A 1-year maturity semi-annual payment bond with a coupon rate of 6% trades at $99.74. What are the 6-month and 1-year zero rates? (For all parts of this question, all interest rates are continuously compounded.)

$$100 = 94e^{R_1(1)}$$
$$R_1 = 6.19\%$$
$$99.74 = 3e^{-R_{.5}(.5)} + 103e^{-.0619(1)}$$
$$R_{0.5} = 5.41\%$$

4.9 B) Without performing any additional calculations, determine the range of values within which the yield on a 1-year maturity semi-annual payment bond should lie.

$R_{0.5} < yield < R_1$, i.e. the yield is "in between" the short and the long zero rates. Thus, 5.41% < yield < 6.19%

4.9 C.) Without performing any additional calculations, what can you infer about the six-month that starts six months from now?

The long zero rate is "in between" the short zero rate and the forward rate, i.e. $R_{0.5} < R_1 < F$. Thus, 6.19% < F.

4.10. Sometime ago, a company entered into an FRA (Forward Rate Agreement), which specifies that the company will receive 7%, quoted with semi-annual compounding, on a principal of $100 million for the 6-month period starting now. The observed 6-month rate equals 8%, quoted with semi-annual compounding. Determine the amount of the settlement, i.e. how much must the company pay or receive now, the start of the FRA period?

The company must pay the bank $480,770.

$$-\$480{,}770 = \frac{100M\,(7\% - 8\%)0.5}{\left[1 + \frac{8\%}{2}\right]}$$

Chapter 5: Determination of Forward and Futures Prices

5.1. An investor shorts 100 shares when the share price is $50 and closes out the position 6 months later when the share price is $43. The shares pay a dividend of $3 per share during the 6 months. What is the investor's profit?

$400 = (50 – 43 – 3) 100

5.2. The spot price of an investment asset that provides no income is $30. The risk-free rate for all maturities is 10% with continuous compounding. What is the 3-year forward price?

$40.50 = 30 e^(.1x3)

5.3. The spot price of an investment asset is $30. The asset provides income of $2 at the end of the 1st year. The asset also provides income of $2 at the end of the 2nd year. There is no additional income generated by the asset during the 3-year life of a forward contract. The risk-free rate for all maturities is 10% with continuous compounding. What is the 3-year forward price?

$35.84 = (30 – 2e^-.1 -2e^.-1x2)e^.1x3

5.4. The spot price of an investment asset that provides no income is $30. The risk-free rate for all maturities is 10% with continuous compounding. What is the value of a long position in a 3-year forward contract where the delivery price is $30?

$7.78 = 30 – 30 e^-.1x3

5.5. A spot exchange rate is $0.7 and the 6-month domestic and foreign risk-free continuously compounded interest rates are 5% and 7%, respectively. What is the 6-month forward rate?

$0.693 = .7 e^(.05-.07)0.5

5.6. A short forward contract with a delivery price of $40 was negotiated sometime ago and will expire in 3 months. The current forward price for a 3-month forward contract is $42. The 3-month risk-free interest rate is 8% with continuous compounding. What is the value of the short forward contract?

-$1.96 = (40-42) e^-.08x.25

5.7. The spot price of an asset is positively correlated with the market portfolio. The current 1-year futures price of the asset is $10. What can you infer about the expected spot price of the same asset a year from now, denoted E(S)?

E(S) > $10. In this situation, normal backwardation prevails.

5.8. The S&P 500 Index has a spot value of $1,095 with a continuously compounded dividend yield of 1%. The continuously compounded interest rate is 5%. What should the 8-month futures price of the index be?

$$F_0 = 1{,}095e^{(5\%-1\%)\frac{8}{12}} = \$1{,}124.60$$

5.9. The spot price of soybeans is $9.80 per bushel. The 9-month futures price of soybeans is $10.20 per bushel. The interest rate and the cost of storage, both quoted as continuously compounded rates, equal 6% and 2%, respectively. Soybeans are considered a consumption good. What is the inferred value of the continuously compounded convenience yield on soybeans?

$$10.2 = 9.8e^{(6\%+2\%-y).75}$$
$$y = 2.7\%$$

5.10. The spot price of rape seed is $19 per bushel. The interest rate, the rape seed cost of storage, and the rape seed convenience yield equal 5%, 1%, and 0.75%, respectively. All rates are expressed as continuously compounded per annum rates. What should be the 6-month futures price of rape seed?

$$F_0 = 19e^{(5\%+1\%-0.75\%).5}$$
$$F_0 = \$19.51$$

Chapter 6: Interest Rate Futures

6.1. A trader enters into a long position in one Eurodollar futures contract. How much does the trader gain when the futures quote increases by 6 basis points?

Gain of $150 = $25x6

6.2. A company invests $1,000 in a 5-year zero-coupon bond and $4,000 in a 10-year zero-coupon bond. What is the portfolio's duration?
9 years = 5(1/5) + 10(4/5)

6.3. In February a company purchases 2 June Eurodollar futures contracts at 95.5. In June the final settlement price of the contract is 97. What has the company accomplished?

In February, the company arranged to lock-in $2 million of investment at 4.5% = (100 – 95.5) % in June. Note that, in the absence of the hedge, the firm would have had to invest at 3% = (100-97) %. After the fact, the hedge was successful in the following specific sense: the firm arranged to invest at an interest rate that turned out after the fact to be high.

6.4. In February a company decides to sell 3 June Eurodollar futures contracts at 95.5. In June the final settlement price of the contract is 97. What has the company accomplished?

In February, the company arranged to lock-in $3 million of financing at 4.5% = (100 – 95.5) % in June. Note that, in the absence of the hedge, the firm would have be able to finance at 3% = (100-97) %. After the fact, the hedge was unsuccessful in the following specific sense: the firm locked-in financing at a rate that turned out to be high after the fact.

6.5. A bond portfolio with a market value of $10 million has a duration of 9 years. The zero curve is flat at 6% per annum compounded continuously. What happens to the market value of the portfolio if interest rates were to rise to 6.5% per annum compounded continuously?

The market value of the portfolio will drop by $450,000, i.e. the change in the value of the portfolio equals -$450,000 = - 9 (.5%) $10M

6.6. A bond portfolio with a market value of $10 million has a duration of 9 years. The zero curve is flat at 6% per annum compounded semiannually. What happens to the market value of the portfolio if interest rates were to rise to 6.5% per annum compounded semiannually?

The market value of the portfolio will drop by $436,893, i.e. the change in the value of the portfolio equals -$436,893 = - {9/ (1 + .06/2)} (.5%) $10M. Note that the modified duration, {9/ (1 + .06/2)}, equals 8.74 years.

Chapter 7: Swaps

Problem 7.1 deals with the post-inception valuation of an interest rate swap. Parts A and B view the value of a swap as the difference between two bonds, one being a fixed rate

bond, and the other being a floating rate bond. Parts C, D and E view the swap as a portfolio of forward contracts, i.e. a portfolio of FRAs. Recall from chapter 4 that an FRA may be valued as if the projected forward rate will prevail.

7.1. The zero curve is flat at 5% per annum with continuous compounding. A swap with a notional principal of $100 in which 6% is received and 6-month LIBOR is paid will last another 15 months. Payments are exchanged every 6 months. The 6-month LIBOR rate at the last reset date, which occurred 3 months ago, was 7%. The company in question receives fixed and pays floating interest rates. What is the value of the swap to the company?

A.) What is the value of the fixed rate bond underlying the swap?

$102.61 = 3 e^-.05x.25 + 3 e^-.05x.75 + 103 e^-.05x1.25

B.) What is the value of the floating rate bond underlying the swap?

$102.21 = (3.5 + 100) e^-.05x.25

3.5 equals .5 x 7% x 100, where 7% is 6-month LIBOR observed 3 months ago; 3.5 is the next interest rate payment that will be paid 3 months from now. The floating rate bond will be worth its par value of 100 immediately after the next interest payment of 3.5.

Since the firm in question receives fixed and pays floating, the value of the swap = $102.61 – $102.21 = $0.4

C.) What is the value of the payment that will be exchanged in 3 months?

-0.49 = (3-3.5) e^-.05x.25

Note that, with regard to part C, there is no uncertainty regarding the cash flows that will be exchanged 3 months from now. All uncertainty was resolved when 6-month LIBOR was observed 3 months ago at a value of 7%.

D.) What is the value of the payment that will be exchanged in 9 months?

.45 = (3-2.5315) e^-.05x.75. The 5% forward rate continuously compounded is first restated as an interest rate with semiannual compounding, i.e., 5.6302%. Thus, 2.5315 = 5.6302% x 100 x .5.

The swap cash flows 9 months from now are viewed as a 9-month FRA.

E.) What is the value of the payment that will be exchanged in 15 months?

.44 = (3-2.5315) e^-.05x1.25. The 5% forward rate continuously compounded is first restated as an interest rate with semiannual compounding, i.e., 5.6302%. Thus, 2.5315 = 5.6302% x 100 x .5.

Viewing the interest rate swap as portfolio of FRAs with staggered maturities, the value of the swap to the company that receives fixed and pays floating equals 0.4 = -.49 + .45 + .44

Note that the two swap interpretations yield consistent results.

7.2. Aussie Pty. Ltd. wishes to borrow USDs (U.S. dollars). Yank Corp. wishes to borrow AUDs (Australian dollars). The following interest rates have been quoted.

Borrowing Firm	Loan in AUDs	Loan in USDs
Aussie Pty. Ltd.	11%	7%
Yank Corp.	10.6%	6.2%

A currency swap has been devised in which Aussie and Yank gain equally. The swap results in Aussie's and Yank's net interest rate liabilities being exclusively in USDs and AUDs, respectively. The bank gains 10 basis points.

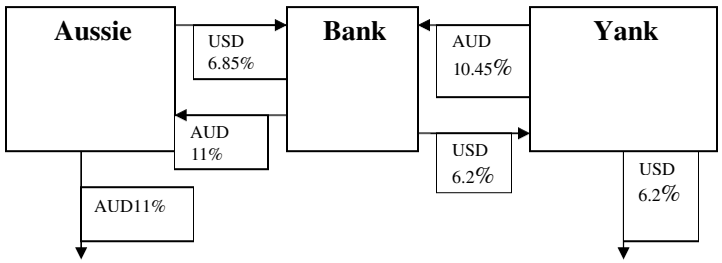
Note that Yank is a higher credit quality firm, enjoying an absolute advantage in both loan types. However, Yank has a comparative advantage in USD debt, whereas Aussie has a comparative advantage in AUD debt. Yank wants AUD debt whereas Aussie wants USD debt. Thus, the preconditions for a mutually beneficial swap are satisfied, i.e. for both swap counterparties the desired type of debt differs from the type of in which comparative advantage is enjoyed.

Interest rate differences	0.4%	0.8%

The total gain is the absolute value of the difference in interest rate differences, i.e. 0.4% or 40 bps. This total gain is partitioned among the parties to the swap. The banks gains 10 bps. The remaining 30 bps is shared equally between Yank and Aussie. Thus Yank and Aussie each gain 15 bps.

A.) What is the USD interest rate that Aussie must pay the bank as part of the swap? Aussie pays the bank USD 6.85%.

Since Yank does not want any liability in USDs, the bank via the swap must compensate Yank for the 6.2% in USD it must pay. Since Aussie does not want any liability in AUDs, the bank via the swap must compensate Aussie for the 11% in AUD it must pay.



B.) What is the AUD interest rate that Yank must pay the bank as part of the swap?
Yank pays the bank AUD 10.45%.

Yank's gain = 10.6% - 10.45% = 0.15% or 15 bps
Aussie's gain = 7% - 6.85 % = 0.15% or 15 bps
Bank's gain = (6.85% - 6.2%) + (10.45% - 11%) = 0.10% or 10 bps

7.3. A $10 million notional principal interest rate swap has a remaining life of 5 months. Under the terms of the swap, 3-month LIBOR is exchanged for 6% per annum (compounded quarterly). The zero or spot rate for all maturities is 4% per annum compounded continuously. The 3-month LIBOR rate was 3.5% per annum (compounded quarterly) a month ago.
A.) What is the value of the floating rate bond implicit in this interest rate swap?

$$Bfloat = (0.0875\ M + 10\ M)e^{-4\%(2/12)} = \$10.0205\ M$$

B.) What is the value of the fixed rate bond implicit in this interest rate swap?

$$Bfix = .15e^{-4\%(2/12)} + 10.15Me^{-4\%(5/12)} = \$10.1312\ M$$

C.) What is the value of the swap to the swap counterparty that receives floating and pays fixed?

Value of swap = $10.0205M - $10.1312 M = -$0.1107 M

Problem 7.4 is an addendum to the boot-strapping procedure for generating the zero or spot curve that was discussed in chapter 4. The new theoretical result that is exploited here is the following: The n-year semi-annual payment swap rate is the n-year par yield on a bond.

7.4. The LIBOR zero rates for 6 months, 1 year, and 18 months equal 5.4%, 5.7%, and 6% continuously compounded, respectively. The swap rate for a 2-year semi-annual payment swap equals 6.6% with semi-annual compounding. What is the 2-year zero rate continuously compounded?

$3.3e^{-5.4\%(.5)} + 3.3e^{-5.7\%(1)} + 3.3e^{-6.6\%(1.5)} + 103.3e^{-R2} = 100$

$e^{-R2} = .8776$

2-year zero rate or R = 6.53%

Problem 7.5 views a currency swap as the difference between two bonds, one denominated in USDs and the other denominated in AUDs. In this case, the company pays in AUDs and receives in USDs. Thus, the value of the swap in USDs is the value of the USD bond minus the value of the AUD bond, with the latter converted into USDs at the current spot rate.

7.5. A currency swap has a remaining life of 9 months, the last exchange of cash flows having occurred 3 months ago. The swap involves a company paying interest at 8% compounded semi-annually on AUD 112 million and receiving interest at 5% compounded semi-annually on USD 100 million every six months. AUD denotes the Australian dollar and USD denotes the U.S. dollar. The zero rates in Australia and the U.S. equal 7% and 4% continuously compounded, respectively, for all maturities. The current exchange rate equals USD 0.95 per AUD. What is the value of the swap, measured in USDs, to the company?

A.) Answer the question interpreting a swap as the difference between two bonds.

$AUD: 112Mx8\%x.5 = AUD4.48M$

$USD: 100Mx5\%x.5 = USD2.5M$

$BAUD = 4.48Me^{-7\%(.25)} + 116.48Me^{-7\%(.75)} = AUD114.925M$

$BUSD = 2.5Me^{-4\%(.25)} + 102.5Me^{-4\%(.75)} = USD101.946M$

$Vswap = BUSD - .95xBAUD = 101.946M - .95(114.925M) = -USD7.233M$

B.) Answer the question interpreting a swap as a portfolio of forward contracts with staggered maturities.

3-month forward:

$F_{.25} = 0.95e^{(4\%-7\%)25} = .9429$

$f_{.25} = \left[USD2.5M - AUD4.48M(.9429)\right]e^{-4\%(.25)} = -USD1.71M$

9-month forward:

$$F_{.75} = 0.95e^{(4\%-7\%).75} = .9289$$

$$f_{.75} = \left[USD102.5M - AUD116.48M(.9289)\right]e^{-4\%(.75)} = -USD5.53M$$

$$V_{SWAP} = f_{.25} + f_{.75} = -USD1.71M - USD5.53M = -USD7.24M$$

Chapter 9: Mechanics of Options Markets

9.1. Consider an exchange traded put option to sell 100 shares for $20. Give the strike price and the number of shares that can be sold after:

A.) A 5 for 1 stock split

$4 =$20/5; 500 = 100x5

B.) A 25% stock dividend

$16=$20/1.25; 125=100x1.25

C.) A $5 cash dividend

$20; 100

9.2. XY Company has 100 million shares outstanding. What happens to that number as a result of each of the following events. Each event should be evaluated separately:

A.)Some exchange traded puts on XY stock are exercised.

B.) Some exchange-traded calls on XY stock are exercised.

C.) Some warrants on XY stock are exercised.

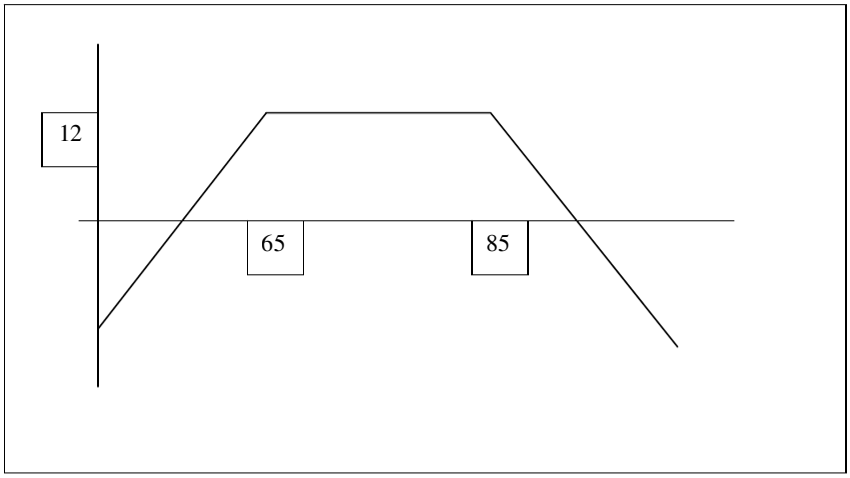
D.) Some bonds convertible to XY stock are converted.

For A. and B. the number of shares outstanding stays equal to 100 million shares. For C. and D. the number of shares outstanding rises above 100 million shares.

9.3. A speculator writes (or sells) a call option with a strike price of $85 and a put option with a strike price of $65 on one share of X Inc. common stock. Both options are European and expire a year from now. The call premium is $7 whereas the put premium is $5. For what values of the yearend stock price will the speculator generate a positive profit?

Option portfolio premium = $12

Positive profit generated for yearend stock price above $(65-12) or $53
and below $(85+12) or $$97.



Chapter 10: Properties of Stock Options

10.1. What is the lower bound for the price of a 2-year European call option on a stock when the stock price is $20, the strike price is $15, the risk-free rate is 5%, and there are no dividends?

$6.43 = 20 – 15(e^-.05x2)

10.2. What is the lower bound for the price of a 2-year European call option on a stock when the stock price is $20, the strike price is $15, and the risk-free rate is 5% and dividends of $1 per share are payable 6 months and 18 months from now?

$4.51 = 20 - 1 e^(-.05x.5) – 1 e^(-.05x1.5) - 15e^(-.05x2)

10.3. What is the lower bound for the price of a 2-year European call option on a stock when the stock price is $20, the strike price is $15, the risk-free rate is 5%, and the continuously compounded dividend yield is 1%?

$6.03 = 20e^(-.01x2) – 15e^(.05x2)

10.4. What is the lower bound for the price of a 2-year European put option on a stock when the stock price is $20, the strike price is $15, the risk-free rate is 5%, and there are no dividends?

0. Note that 15(e^-.05x2)- 20 = -$6.43 but any option cannot have a negative value

10.5. What is the lower bound for the price of a 6-month European put option on a stock when the stock price is $40, the strike price is $46, the risk-free interest rate is 6% and there are no dividends?

$4.64 = 46e^(-.06x.5) – 40

10.6. What is the lower bound for the price of a 6-month European put option on a stock when the stock price is $40, the strike price is $46, the risk-free interest rate is 6% and dividends per share of $2 are payable 3 months from now?

$6.61 = 46e^(-.06x.5) – (40 – 2e^(-.06x.25))

10.7. What is the lower bound for the price of a 6-month European put option on a stock when the stock price is $40, the strike price is $46, the risk-free interest rate is 6% and the continuously compounded dividend yield is 2%?

$5.04 = 46e^(-.06x.5) – 40e^(-.02x.5)

10.8. The price of a European call option on a non-dividend paying stock with a strike price of $50 is $6. The stock price is $51, the risk-free interest rate is 6% and the time to maturity is 1 year. What is the price of a 1-year European put option on the stock with a strike price of $50?

$2.09 since 6-P = 51 – 50e^(-.06) implies P = 2.09

10.9. The price of a European call option on a stock, which will pay a dividend per share of $1 3 months from now, is $6. The strike price is $50. The stock price is $51, the risk-free interest rate is 6% and the time to maturity is 1 year. What is the price of a 1-year European put option on the stock?

$3.07 since 6-P = 51 - 1 e^-(.06x.25) – 50 e^(-.06) implies P = 3.07

10.10. The price of a European call option on a stock, which pays a continuously compounded dividend yield of 2%, is $6. The strike price is $50. The stock price is $51, the risk-free interest rate is 6% and the time to maturity is 1 year. What is the price of a 1-year European put option on the stock?

$3.10 since 6-P = 51e^(-.02) – 50e^(-.06) implies P = 3.10

10.11. A call and a put on a stock have the same strike price and time to maturity. Both options are European. At 11AM on a certain day, the price of the call is $3 and the price of the put is $4. At 11:01 AM news reaches the market that results in an increase in the volatility of the stock with no additional effects on either the stock price or the risk-free interest rate. The price of the call option rises to $4.50. What would you expect the price of the put to change to?

$5.50 since 4.50 –P = -1 = S – Xe^(-RxT) implies P = 5.50

10.12. The exercise price of a European put option on a single stock, which is currently trading at $45 per share, is $50. The sole dividend per share envisioned during the 6-month life of the option is $2 to be paid 3 months from now. The interest rate is 6% continuously compounded and the put premium equals $4. Specify associated dollar amounts in your answers to the following questions:
A.) What transactions now will generate arbitrage profits?

The lower bound for the put premium is violated:

$$p \geq Ke^{-rt} - (S - D)$$

$$4 \geq ?50e^{-6\%(.5)} - (45 - 1.97)$$

$$4 < 5.49$$

Gap = 5.49 – 4 = $1.49 is the arbitrage profit that can be generated now.

Buy put	-$4
Borrow	$48.52 = 50 e^-(6%x0.5)
Borrow	$1.97 = 2 e^-(6%x0.25) Total amount borrowed = 50.49
Buy stock	-$45
Profit =	$1.49

Note: After 3 months, use dividend received of $2 to pay off borrowing $1.97.

B.) What transactions 6 months from now will generate arbitrage profits?

If St < $50	
Exercise put, obtain	$50
Payoff loan of $48.52	-50
Profit =	0

If St > $50	
Allow put to lapse unexercised	
Sell stock	St
Payoff loan of $48.52	-50
Profit =	(St – 50)

Chapter 11: Trading Strategies Involving Options

11.1 6-month European call options with strike prices of $35 and $40 cost $6 and $4, respectively.

A.) What is the maximum gain or profit when a bull spread is created from the calls?

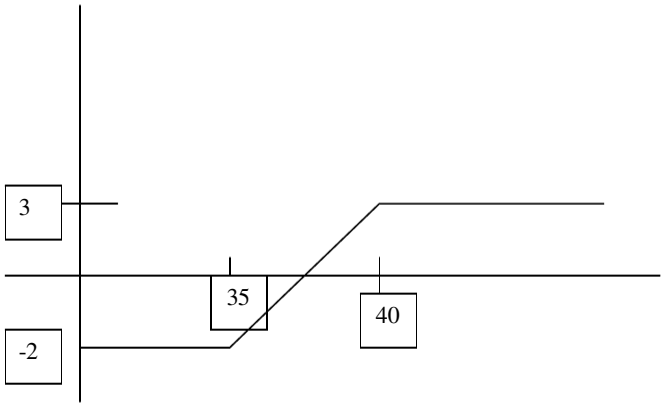
$3. See graph below.

B.) What is the maximum loss (negative profit) when a bull spread is created from the calls?

$2. See graph below.

C.) Under what conditions regarding P, the stock price 6 months from now, will profits be generated from the indicated bull spread?

When P exceeds $37. See graph below.



D.) What is the maximum gain or profit when a bear spread is created from the calls?

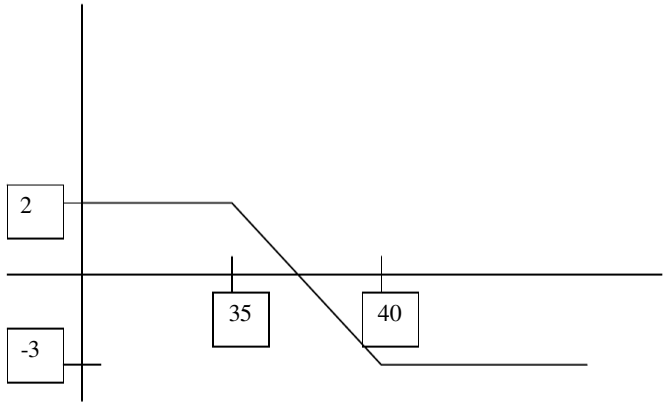
$2. See graph below.

E.) What is the maximum loss (negative profit) when a bear spread is created from the calls?

$3. See graph below.

F.) Under what conditions regarding P, the stock price 6 months from now, will profits be generated from the indicated bear spread?

When P is less than $37. See graph below.



11.2 6-month European put options with strike prices of $55 and $65 cost $8 and $10, respectively.

A.) What is the maximum gain when a bull spread is created from the puts?

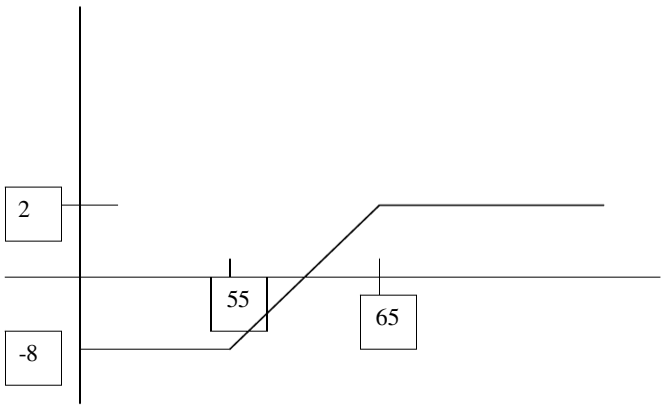
$2. See graph below.

B.) What is the maximum loss when a bull spread is created from the puts?

$8. See graph below.

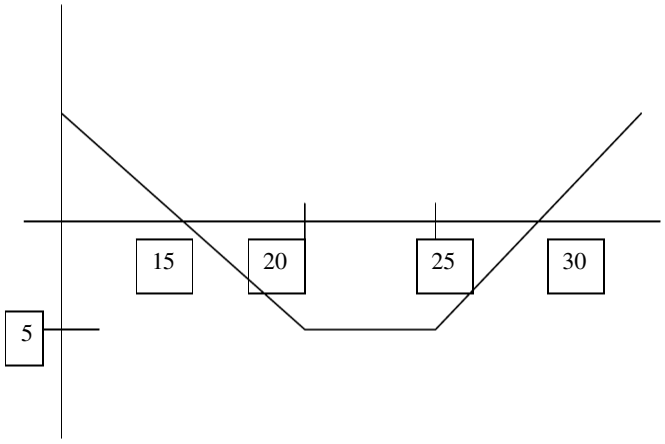
C.) Under what conditions regarding P, the stock price 6 months from now, will profits be generated from the indicated bull spread?

When the stock price 6 months from now exceeds $63. See graph below.



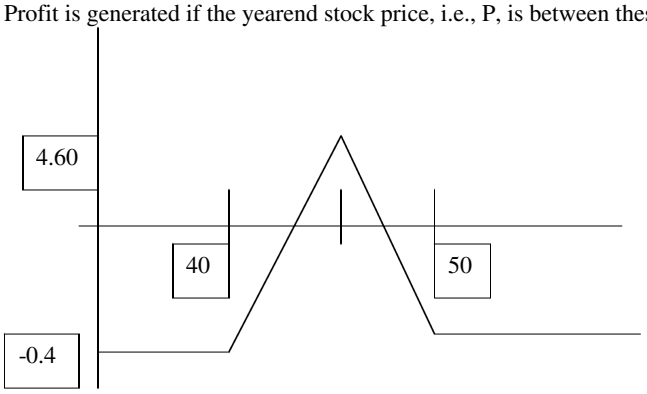
11.3. A 3-month call with a strike price of $25 costs $2. A 3-month put with a strike pride of $20 costs $3. A trader uses the options to create a strangle. For what 2 values of the stock price 3 months from now will the trader breakeven?

$15 and $30. See graph below.



11.4. A speculator decides to create a butterfly spread involving the following 3 one-year European call options on a stock with exercise prices (and corresponding call premia in parentheses): $40 (premium $3), $45 (premium $2.30) and $50 (premium $2). For what values of the yearend stock price, denoted P, will profits be generated?

Profit if $40.40 < P < $49.60
The first horizontal intercept occurs at $40 + $0.4 = $40.40
The second horizontal intercept occurs at $50 -$0.4 = $49.60
Profit is generated if the yearend stock price, i.e., P, is between these two values.

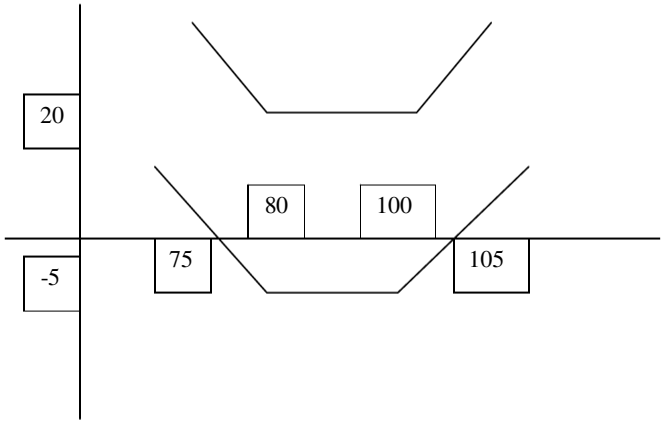


The following is an example of what I refer to in the slides as a **strangle not!**

11.5. A speculator purchases a put option with exercise price of $100 and a premium of $15 and a call option with exercise price of $80 and a premium of $10. The options are European, involve one share in Y Corp., and expire a year from now. For what values of the yearend stock price will the speculator generate a positive profit?

Profit generated if yearend stock price is less than $75 or greater than $105.

In the following graph, the up-front portfolio premium equals $25. The yearend stock price is plotted on the horizontal axis. The upper graph depicts the payoff diagram whereas the bottom graph depicts the profit diagram. Recall that profit = payoff – upfront premium.



Chapter 12: Introduction to Binomial Trees

12.1. Consider a 6-month European put option on a non-dividend paying stock with a strike price of $32. The current stock price is $30 and over the next 6 months it is expected to rise to $36 or fall to $27. The risk-free interest rate is 6%.

A.) What is the risk-neutral probability of the stock rising to $36?

0.435 = (1.0305 - .9) / (1.2 - .9) where u = 1.2; d=.9; e^RT = 1.0305

B.) What position in the stock is necessary to hedge a long position in 1 put option?

Long position or own 0.556 share. Delta = (0 – 5) / (36 – 27) = -.556. Delta is the number of shares that must be owned to hedge a short position in a put option. Since we are trying to hedge a long position in one put, the appropriate position in the stock is .556.

C.) What is the value of a put option?

$2.74 via risk-neutral valuation. 2.74 = (e^-.06x.5) [.565x5]

D.) Assume now that the option is a call option rather than a put option. What position in the stock is necessary to hedge a long position in 1 call option?

Short position or issue 0.444 share. Delta = (4 - 0) / (36 – 27) = .444. If you issue a call option, you hedge by owning .444 share. Thus, if you own a call option, you must short .444 share.

E.) What is the value of a call option?

$1.69 Via risk-neutral valuation 1.69 = (e^-.06X.5) [.435x4]. Via put-call-parity C = 2.74 + 30 – 32(e^-.06x.5) = 1.69

12.2. A power option pays off [max(St – K), 0]^2 at time t where St is the stock price at time t and K is the strike price. Note: Since the indicated payoff is merely the squared value of the payoff on a traditional call option, a power option may be considered a call option on steroids. Consider a situation where K=26 and t is one year. The stock price is currently $24 and at the end of one year, it will be either $30 or $18. The risk-free rate is 5%.

A.) What is the risk-neutral probability of the stock rising to $30?

0.603 = (1.05127 - .75) / (1.25 - .75) where u = 1.25; d = .75; e^Rt = 1.05127

B.) What position in the stock is required to hedge a short position in one power option?

Long position in 1.333 shares. Delta = (16 - 0) / (30-18) = 1.333.

C.) What is the value of the power option?

$9.17 = (e^-.05) [.6025x16] via risk-neutral valuation.

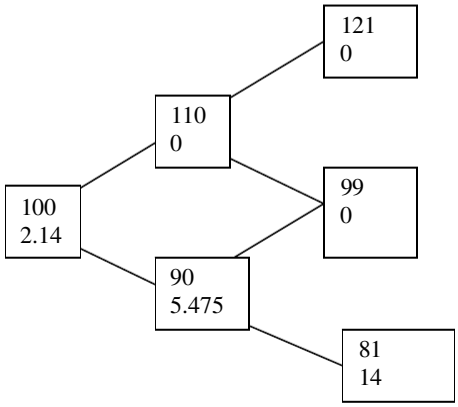
12.3. A stock price is currently $100. The stock pays no dividend. Over each of the next two 3-month periods, it is expected to increase by 10% or fall by 10%; the preceding percentages are not annualized. Consider a 6-month European put option with a strike price of $95. The risk-free rate is 8%.

A.) What is the risk-neutral probability of a 10% rise in a single quarter?

.601 = (1.0202 - .9) / (1.1 - .9)

B.) What is the value of the option?

$2.14. Refer to the following diagram. Following the notational convention in the textbook, the upper number at each node refers to the value of the stock and the lower number refers to the value of the option.

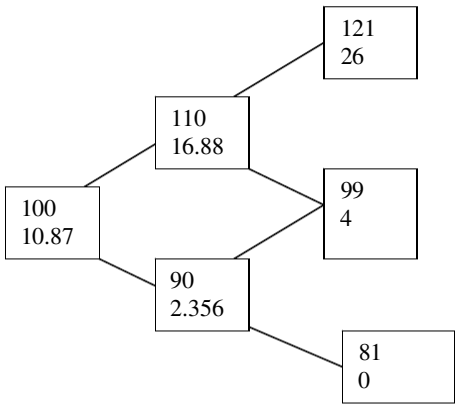


C.) If the put option were American rather than European, what would its value be?

$2.14. Inspection of the 3-month or intermediate nodes shows that it would never be optimal to prematurely exercise the put. Thus, in this case, the value of the American put equals the value of the otherwise identical European put.

D.) Assume now that the option is a European call rather than a European put. What is the value of the option?

$10.87. Refer to the following diagram.



An alternative way of valuing the call option is to invoke put-call parity, C = 10.87 = 2.14 +100 – 95 e^(-.08x.5).

Note that the call option values at the end of one quarter, i.e. 16.88 and 2.356, are obtained via risk-neutral valuation. Thus, for example, 2.356 = e^(-.08x.25) [.601x4].

E.) Assume now that the call option is American rather than European. What is the value of the option?

$10.87. It is never optimal to prematurely exercise an American call option on a non-dividend paying stock. Thus, the value of this American call option equals the value of an otherwise identical European call option.

12.4. A common stock that pays no dividend has a price that currently equals $50. At the end of 3 months the stock price can either rise to $54 or drop to $47. The risk-free interest rate is 10% per annum compounded continuously. Consider a 3-month European call option on 100 shares with a strike price of $49. To hedge the writing of such an option, what position must be taken in the underlying stock?

$$\Delta = \frac{5-0}{54-47} = .714 = .71$$

Must purchase or take a long position in 71 shares of the underlying stock.

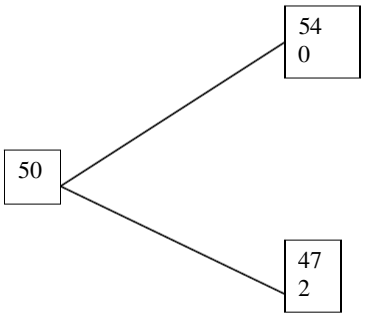
Note: Call value in up state (stock price = 54) is 5. Call value in down state (stock price = 47) is 0.

12.5. A common stock that pays no dividend has a price that currently equals $50. At the end of 3 months the stock price can either rise to $54 or drop to $47. The risk-free interest rate is 10% per annum compounded continuously. Consider a 3-month European put option on 100 shares with a strike price of $49. To hedge the writing of such an option, what position must be taken in the underlying stock?

T = .25, r = 10%, K = 49

$$\Delta = \frac{f_u - f_d}{Su - Sd} = \frac{0-2}{54-47} = -.2857 \approx -.29$$

Must take a long position in -29 share or a short position in 29 shares.



Chapter 13: Valuing Stock Options: The Black-Scholes-Merton Model

13.1. For a European call option on a non-dividend-paying stock, the stock price is $30, the strike price is $29, the risk-free interest rate is 6%, the volatility is 20% per annum and the time to maturity is 3 months. Express you answers in terms of the N(d), i.e., calculate d but do not calculate N(d).

A.) What is the price of the option?

30N(.539) – 28.57 N(.439) since c = 30 N(d1) – 29 e^(-.06x.25) N(d2) where
d1 = (ln(30/29) + (.06 + (.2^2)/2).25 / (.2 (.25)^.5) and d2 = d1 - (.2 (.25)^.5)

B.) What is the price of the option if it were American rather than European?

Same as part A.) since it is never optimal to prematurely exercise an American call option on a non-dividend paying stock.

C.) What is the price of the option is if it is a put?

28.57N(-.439) – 30N(-.539)

D.) What is the price of the option if a dividend of $2 is expected in 2 months?

28.02N(-.1438) – 28.57 N(-.2438) since in part A.) instead of the stock price of $30 we substitute the stock price minus the present value of the dividend, i.e. 30 – 2e^(-.06x.1666) = $28.02

13.2. The underlying stock of a 6-month American call option will pay a dividend at the end of 5 months. The strike price is $30 and the risk-free rate is 10%. How high must the dividend per share be for there to be some chance of early exercise?

$0.25 = 30 (.1) ((6-5)/12) Note: This formula is the last inequality found in the Appendix to Chapter 13 on page 313.

The dividend per share must exceed $0.25 for there to be a possibility of optimal exercise of the call option immediately before the ex-dividend date 5 months from now, i.e. early exercise.

The following is an explanation of the formula:

There is no chance of early exercise if the DPS (dividend per share) is less than a certain critical value. In this situation, you can rule out the possibility that the option will be exercised early.

That critical value is calculated as follows: strike price x risk-free rate x time span between the date of the last dividend payment and the expiration date of the option. Time is measured in years.

If the DPS exceeds the critical value, there is some chance that the call will be exercised early, i.e., one cannot rule out the possibility of early exercise.

Early exercise means that the call option will be exercised before the expiry of the option.

13.3. A call option on one share has one year to expiration and stipulates an exercise price of $60. The underlying stock is currently trading at $50 per share, exhibits a volatility of 30%, and pays no dividend. The risk-free interest rate is 6% continuously compounded. The option is European.

A.) What is the risk neutral probability that the option will be exercised?

$$d_2 = \frac{\ln(\frac{50}{60}) + (.06 - \frac{.3^2}{2})1}{.3\sqrt{1}} = -.5577$$

$$N(-.5577)$$

B.) If a hedge fund were to write a call option on 100 shares of the underlying common stock, what position must the fund take in the underlying stock to form a riskless or arbitrage portfolio?

$$d_1 = d_2 + \sigma\sqrt{T}$$

$$d_1 = -.5577 + .3\sqrt{1} = -.2577$$

$$N(d_1) = N(-.2577)$$

The hedge fund must hold a long position in 100N(-.2577) shares of the underlying common stock.

13.4. A European put option that expires in 3 months stipulates a strike price of $70 per share. The underlying common stock is currently trading at $75 per share and exhibits a volatility of 35%. The risk-free interest rate is 5% continuously compounded. The only dividend that will be paid during the life of the option is $3 per share that is payable two months from now.

A.) What is the risk neutral probability that the put option will be exercised?

$$D = 3e^{-.05(2/12)} = 2.975$$

$$d_2 = \frac{\ln(\frac{75 - 2.975}{70}) + \left(.05 - \frac{.35^2}{2}\right).25}{.35\sqrt{.25}} = 0.1468$$

$$N(-d_2) = N(-.1468) \approx .44$$

Note: In both the quizzes and the final exam, you will not be asked to read probabilities off a normal probability table. In this specific case, the required answer will be N(-0.1468), not approximately 0.44.

B.) If a hedge fund were to write a put option on 100 shares of the common stock, want position in the common stock must the hedge fund establish to form a riskless or arbitrage portfolio?

$$d_1 = d_2 + \sigma\sqrt{T}$$

$$d_1 = .1468 + .35\sqrt{.25} = .3218$$

$$N(-d_1) = N(-.3218)$$

Recall that the delta of a put on one common stock equals –N(-d1). Thus, to hedge the issuance of a put on one common stock, the hedge fund must take a long position in –N(-d1) shares of the underlying stock, i.e., the fund must take a short position in N(-d1) shares of the underlying stock.

Thus, the hedge fund must take a short position in 100N(-.3218) shares of the underlying common stock.

Chapter 15: Options on Stock Indices and Currencies

15.1. Consider a European put option on a stock index. The index level is 1,000, the strike price is 1,050, the time to maturity is 6 months, the risk-free rate is 4% and the dividend yield on the index is 2%. What is the lower bound to the option price?

$39.16 = 1050e^(-.04x.5) – 1000e^(-.02x.5)

15.2. Consider a European call option on a stock index. The index level is 1,000, the strike price is 900, the time to maturity is 6 months, the risk-free rate is 4% and the dividend yield on the index is 2%. What is the lower bound to the option price?

$107.87 = 1000e^(-.02x.5) – 900 e^(-.04x.5)

15.3. Consider a 1-year European call option on a currency. The exchange rate is $1.0000, the strike price is $0.9100, the domestic risk-free rate is 5%and the foreign risk-free rate is 3%. What is the lower bound to the option price?

$0.1048 = 1.000e^(-.03) – 0.9100e^(-.05)

15.4. An exchange rate is currently $0.8. It is expected to move up to $0.84 or down to $0.76 in the next 3 months. The risk-free rates in the domestic and foreign currencies are 4% and 6%, respectively.

A.) What is the probability of an up movement in a risk-neutral world?

0.4501 = (e^(.04-.06)x.25 - .95) / (1.05 - .95) where u = 1.05 and d = .95

B.) What is the value of a 3-month European call option with a strike price of $0.82?

$.0089 = e^(-.04x.25) [.4501 (.02)]

C.) What is the value of a 3-month European put option with a strike price of $0.82?

$.0327 = e^(-.04x.25) [.5499 (.06)] where the probability of a down movement = .5499

15.5. A stock index currently equals 1,000. Its volatility is 20%. The risk-free rate is 4% and the dividend yield on the index is 2%. Express you answers in terms of N(d), i.e. calculate d but do not calculate N(d).

A.) What is the value of a 1-year European call option with a strike price of 950?

980.20 N(.4565) – 912.75 N(.2565) = 1000e^(-.02) N(d1) – 950 e^(-.04) N(d2)
Where d1 = (ln(1000/950) + (.04 - .02 + .2^2/2)) / .2 and d2 = d1 - .2

B.) What is the value of a 1-year European put option with a strike price of 950?

912.75 N(-.2565) – 980.2 N(-.4565) = 950 e^(-.04) N(-d2) - 1000e^(-.02) N(-d1)

15.6. A portfolio manager in charge of a portfolio worth $10 million is concerned that the market might decline rapidly during the next 6 months and would like to use options on the S&P 100 to provide protection against the portfolio falling below $9.5 million. The S&P 100 index is currently standing at 500 and each contract is on 100 times the index. Assume dividend yields equal zero on both the portfolio and the stock index.

A.) If the portfolio has a beta of 1, how many put option contracts should be purchased?

200 = 1 (10M/500x100)

B.) If the portfolio has a beta of 1, what should be the strike price of the put options?

475 since [9.5/10 - 1 – R] = 1 [K/500 - 1 – R] implies K = 475. R is the redundant 6-month risk-free rate. Note that R is redundant if and only if the beta of the portfolio equals 1, because R cancels out from both sides of the equation.

C.) If the portfolio has a beta of 0.5, how many put option contracts should be purchased?

100 = .5 (10M/500x100)

D.) If the portfolio has a beta of 0.5, what should be the strike price of the put option? Assume that the risk-free rate is 10% and the dividend yield on both the portfolio and the index is 2%, where both interest rates and dividend yields are quoted with semi-annual compounding.

430 since [(9.5/10 -1) +.01 - .05] = .5 [(K/500 -1) +.01 - .05] implies K = 430

15.7. A stock portfolio, whose beta equals 1.8, is worth $50 million and the S&P 500 Index is at 1,350. The dividend yield on both the portfolio and the index equals 2.5% while the risk-free interest rate equals 6%, both numbers expressed with annual compounding. How would you ensure that the yearend ex-dividends value of the portfolio not fall below $43 million?

$$NumberPuts = 1.8\frac{50M}{1350(100)} = 667Puts$$

$$\left[\frac{43M}{50M} - 1 + .025 - .06\right] = 1.8\left[\frac{K}{1350} - 1 + .025 - .06\right]$$

$$K = 1266$$

Purchase 667 puts with strike price of $1,266.